

The lingual articulation of devoiced /u/ in Tokyo Japanese

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Abstract

In Tokyo Japanese, /u/ is devoiced between two voiceless consonants. Whether the lingual vowel gesture is present in devoiced vowels remains an open debate, largely because relevant articulatory data has not been available. We report ElectroMagnetic Articulography (EMA) data that addresses this question. We analyzed both the trajectory of the tongue dorsum across VC1uC2V sequences as well as the timing of C1 and C2. These analyses provide converging evidence that /u/ in devoicing contexts is optionally targetless—the lingual gesture is either categorically present or absent but not reduced. When present, the magnitude of the lingual gesture in devoiced /u/ is comparable to voiced vowel counterparts. Individual speakers varied in how often they produced a word with targetless /u/, but there were consistent patterns across speakers in the environments that were more/less likely to favor targetless /u/. Tokens lacking a lingual vowel target also showed differences consonant timing, which can be modelled as a shift from C-V to C-C coordination.

Keyword: Japanese, devoicing, articulatory phonetics, EMA, linear interpolation, Discrete Cosine Transform (DCT), Bayesian classifier, gestural coordination

General background

This paper examines the lingual articulation of devoiced /u/ in Tokyo Japanese. A classic description of the devoicing phenomenon is that "high vowels are devoiced between two voiceless consonants and after a voiceless consonant before a pause" (Fujimoto, 2015; Kondo, 1997, 2005; Tsuchida, 1997 among many others). This sort of description applies to vowels in numerous other languages (Greek, Shanghai Chinese, Korean, Montreal French: (Jun, Beckman, & Lee, 1998), but Tokyo Japanese is arguably the best studied case of vowel devoicing.

There is a large body of work on this phenomenon in Japanese, covering its phonological conditions (e.g., Kondo, 2005; Tsuchida, 1997), its interaction with other phonological phenomena like pitch accent (e.g., Maekawa, 1990; Maekawa & Kikuchi, 2005; Vance, 1987), its acoustic and perceptual characteristics (Beckman & Shoji, 1984; Faber & Vance, 2000; Matsui, 2014; Nielson, 2015), and studies of the vocal folds (Fujimoto, Murano, Niimi, & Kiritani, 2002; Hirose, 1971; Tsuchida, 1997). See Fujimoto (2015) a recent, comprehensive overview of this research tradition. While now we have a good understanding of many aspects of high vowel devoicing in Tokyo Japanese, what is understudied is the lingual gestures of high vowels when they are devoiced. The only study that we are aware of is Funatsu & Fujimoto (2011), which used EMMA (ElectroMagnetic Midsagittal Articulography) and found little differences between devoiced and voiced /i/ in terms of lingual articulation. However, this

experiment used only one speaker and one item pair (/kide/ vs. /kite/). The study included four repetitions of each item, and offered no quantitative analyses of the data. It is thus probably safe to say that lingual characteristics of devoiced vowels are still understudied—our study is intended to fill this gap.

Why is it important to study the lingual gestures of devoiced vowels? There are a few motivations behind the current study. First, consider Figure 1, taken from Fujimoto et al.'s (2002) study on the glottal gestures of high vowel devoicing in Japanese.

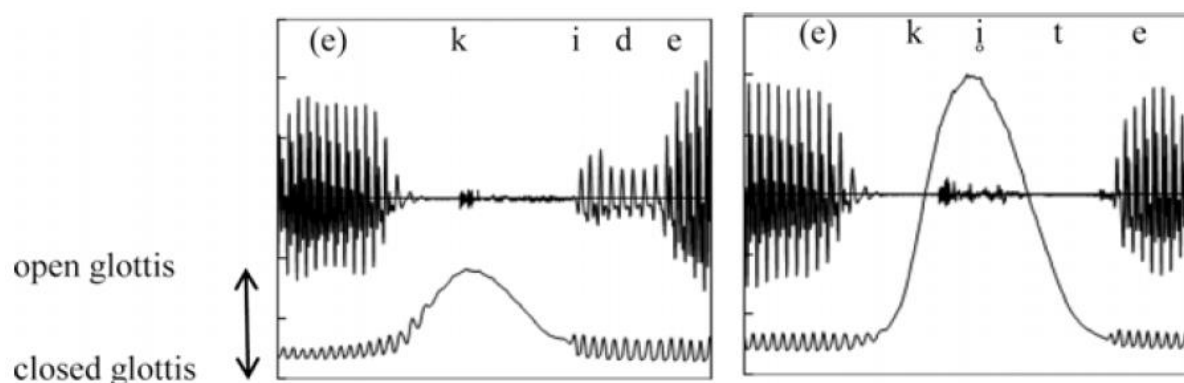


Figure 1: The degrees of glottal abduction in Japanese. The left panel: a voiceless stop followed by a voiced stop, which has a single abduction gesture of /k/. The right panel: a voiceless stop /k/ followed by a voiceless vowel and another voiceless stop /t/. Taken from Fujimoto et al. (2002), cited and discussed in Fujimoto (2015).

Figure 1 shows that a Japanese devoiced vowel shows a single laryngeal gesture of greater magnitude than a single consonant gesture, or even sum of two voiceless consonant gestures (c.f., Munhall & Lofqvist, 1992 for English which shows the latter pattern). This observation

1 implies that Japanese devoiced vowels involve active laryngeal abduction, not simply an
2 overlap of two surrounding gestures. This conclusion in turn implies that Japanese speakers
3 exert *active* laryngeal control over devoiced high vowels (cf. Jun & Beckman 1993 for an
4 analysis that relies on passive gestural overlap, to be discussed below). To the extent that
5 Japanese speakers actively control the laryngeal gesture for devoiced vowels, are lingual
6 gestures of devoiced vowels also actively controlled? and if so, how? In order to answer these
7 questions, this paper tests four specific hypotheses about the lingual gestures of devoiced
8 vowels, to be formulated below in (1).

9
10 The second question that drove our research is whether “devoiced” vowels are simply devoiced
11 or deleted. This issue has been discussed extensively in previous studies of Japanese high vowel
12 devoicing. Kawakami (1977: 24-26) argues that vowels delete in some environment and
13 devoice in others, but he offers no phonological or phonetic evidence. Vance (1987) raised and
14 rejected the hypothesis that high vowels in devoicing contexts are deleted. Kondo (2001) argues
15 that high vowel devoicing is actually deletion based on a phonological consideration. Devoicing
16 in consecutive syllables is often prohibited (although there is much variability: Nielsen 2015),
17 and Kondo argues that this prohibition stems from a prohibition against complex onset or
18 complex coda (i.e. *CCC). On the other hand, Tsuchida (1997) and Kawahara (2015) argue that
19 bimoraic foot-based truncation (Poser, 1990) counts a voiceless vowel as one mora (e.g. [suto]
20 from [sutoraiki] ‘strike’, *[stora]). If /u/ was completely deleted losing its mora, the bimoraic
21 truncation should result in *[stora]. Hirayama (2009) makes a similar phonological argument
22 by showing that devoiced vowels' moras are just as relevant for Japanese *haiku* poetry as voiced
23 vowels' moras. However, just because moras for the devoiced vowels remain, it does not mean
24 that the remaining consonant can host that mora and syllable—this hypothesis is actually
25 proposed by Matsui (2014), who argues that Japanese has consonantal syllables in this

environment. Thus, evidence for either deletion or devoicing from a phonological perspective is mixed (see Fujimoto 2015: 197-198 for other studies on this debate).¹

Previous acoustic studies show that on spectrograms, vowels leave no trace of lingual articulation except for coarticulation on surrounding consonants, which lead them to conclude that vowels are deleted (Beckman 1982; Beckman & Shoji 1984; Whang 2014). Beckman (1982: 118, footnote 3) states that "deletion" is a better term physically, because "there is generally no spectral evidence for a voiceless vowel", whereas "devoicing" is a better term psychologically, because Japanese speakers hear a voiceless vowel even in the absence of spectral evidence (c.f., Dupoux, Kakehi, Hirose, Pallier, & Mehler, 1999). This statement embraces the "deletion" view, at least at the speech production level. Even if vowel devoicing involves phonological deletion, it could be the case that its application is optional or variable, influenced by various linguistic and sociological factors (Fujimoto, 2015; Nielsen, 2015).

Not all phonetic studies have embraced this deletion view, however. The clearest instantiation of this view is "gestural overlap theory" of high vowel devoicing (Faber & Vance, 2000; Jun & Beckman, 1993; Jun et al., 1998). In this theory, high vowel devoicing occurs when a spread glottis gesture of the surrounding consonants overlap with the vowel's glottal gesture (though cf. Figure 1). In this sense, the high vowel devoicing processes in Japanese (and other languages like Korean) are "not...phonological rules, but the result of extreme overlap and hiding of the vowel's glottal gesture by the consonant's gesture" (Jun & Beckman 1983: p.4). This theory

¹ Tsuchida (1997) argues that there is "phonological devoicing" and "phonetic devoicing".

implies that there is actually no deletion—oral gestures remain the same, but do not leave their acoustic traces because of devoicing.

To summarize, there is an active debate about whether the lingual gestures of “devoiced” vowels in Japanese are present but inaudible due to devoicing or absent altogether, possibly because of phonological deletion. Vance (2008), the most recent and comprehensive phonetic textbook on Japanese, states that this issue is not yet settled. Studying lingual gestures of devoiced vowels will provide the crucial new evidence to resolve the debate. Recall that the only past study on this topic, Funatsu & Fujimoto (2011), is based on a small number of tokens and one speaker. In this paper, we expand the empirical base, reporting more repetitions (10-15) of ten real words produced by six naive speakers, and we deploy rigorous quantitative methods of analysis, as detailed by Shaw and Kawahara (submitted). While Shaw and Kawahara (submitted) focused on motivating the computational methodology, this paper offers more detailed reports of the lingual movements of Japanese devoiced vowels, including both the target vowel and the flanking consonants.

Four specific hypotheses tested

Building on the previous studies reviewed in this section, we entertain four specific hypotheses about the lingual articulation of devoiced vowels, stated in (1)

(1) Hypotheses about the status of lingual articulation in devoiced vowels

H1: **full lingual targets**—the lingual articulation of devoiced vowels is the same as for voiced counterparts.

H2: **reduced lingual targets**—the lingual articulation of devoiced vowels is phonetically reduced relative to voiced counterparts.

1 H3: **targetless**—devoiced vowels have no lingual articulatory target.²

2 H4: **optional target**—devoiced vowels are sometimes targetless (deletion is
3 optional, token-by-token).

4
5 The passive devoicing hypothesis, or the gestural overlap theory (e.g. Jun & Beckman 1993),
6 maintains that there is actually no phonological deletion, and hence would predict that lingual
7 gestures would remain intact (=H1). This is much like Brownman and Goldstein's (1992)
8 argument that apparently deleted [t] in *perfect memory* in English keeps its tongue tip gesture.
9 This is also the conclusion that Funatsu & Fujimoto (2011) reached for devoiced /i/ in their
10 sample of EMMA data. Besides the small sample size mentioned above, another caveat is that
11 the study collected nasal endoscopy data to image the vocal folds concurrently with the EMMA
12 data, which may have promoted stable lingual gestures across contexts, since retraction of the
13 tongue body may trigger a gag-reflex.

14
15 Even if devoiced high vowels are not phonologically deleted, it would not be too surprising if
16 the lingual gestures of high vowels were phonetically reduced, hence H2 in (1). At least in
17 English, less informative segments tend to be phonetically reduced (Aylett & Turk, 2004, 2006;
18 Bell, Brenier, Gregory, Girand, & Jurafsky, 2009; Jurafsky, Bell, Gregory, & Raymond, 2001).
19 In Japanese, the segmental identity of devoiced vowels is often highly predictable from context

² We use the term “targetless” rather than “deletion”, because the latter terms commits to (1) a surface representation, (2) a process mapping an underlying representation to a surface representation and (3) the identity of the underlying representation. Our experiment is solely about the surface representation, and hence the term “targetless” is better.

(Beckman & Shoji, 1984; Whang, 2014). Due to devoicing, moreover, the acoustic consequences of a reduced lingual gesture would not be particularly audible to listeners. Hence, from the standpoint of effort-distinctiveness tradeoff, it would not be surprising to observe reduction of oral gestures in high devoiced vowels. Why should speakers articulate vowels that are predictable and not audible to listeners?

The phonological deletion hypothesis, which was proposed by various authors reviewed above, predicts that there should be no lingual targets for devoiced vowels (=H3 in (1)). However, we know that many if not all phonological patterns are variable, i.e., optional (e.g., Coetzee & Pater, 2011). Therefore, we need to consider the possibility that deletion of lingual gestures in devoiced vowels is optional, token-by-token (=H4). The method that we deploy below (Shaw & Kawahara, submitted) will allow us to access both inter-speaker variability as well as within-speaker variability.

Our experiment was designed to test these four hypotheses.

Methodology

Speakers

Six native speakers of Tokyo Japanese (3 male) participated. Participants were aged between 19 and 22 years at the time of the study. They were all born in Tokyo, lived there at the time of their participation in the study, and had spent no more than 3 months outside of the Tokyo region. Procedures were explained to participants in Japanese by a research assistant, who was also a native speaker of Tokyo Japanese. All participants were naïve to the purpose of the experiment. They were compensated for their time and local travel expenses.

1

2 **Materials**

3 A total of 10 target words, listed in Table 1, were included in the experiment. These included
 4 five words containing /u/ in a devoicing context (second column) and a set of five corresponding
 5 words with /u/ in a voiced context (third column). The target /u/ is underlined in each word.
 6 Together, these 10 words constitute minimal pairs or near minimal pairs. The word pairs are
 7 matched on the consonant that precede /u/. They differ in the voicing specification of the
 8 consonant following /u/. The consonant following /u/ is voiceless in devoicing context words
 9 (second column) and voiced in voiced context words (third column). This study focused on /u/
 10 and did not consider /i/ for several practical reasons, the most important one being that
 11 collecting both /u/ and /i/ tokens would mean reducing the repetitions for each target word and
 12 the analytical approach we planned (following Shaw and Kawahara, submitted) requires a large
 13 number of repetitions per word. We focused on /u/ rather than /i/, because the former is more
 14 likely to be devoiced, as confirmed by the study of high vowel devoicing using the Corpus of
 15 Spontaneous Japanese (Maekawa & Kikuchi 2005; see also Fujimoto et al. 2015 and references
 16 cited therein).

17

18 **Table 1: stimulus items. W and K show the environments in which Kawakami (1971)**
 19 **and Whang (2014) predict deletion.**

Comments	Devoicing/deletion (singleton controls)	Voiced vowel (singleton controls)
V deletion (K, W)	ϕ <u>u</u> soku 不足	ϕ <u>u</u> zoku 付属 ‘enclosed’
V devoicing (K, W)	ʃ <u>u</u> taisei 主体性	ʃ <u>u</u> daika 主題歌 ‘theme song’
V deletion (K, W)	kats <u>u</u> toki 勝つ時	kats <u>u</u> dou 活動 ‘activity’
V devoicing (K) V deletion (W)	Hak <u>u</u> sai 白菜	yak <u>u</u> zai 薬剤
V deletion (K, W)	mas <u>u</u> taa マスター	mas <u>u</u> da 益田

1
2 The consonant preceding /u/ draws from the set: /s/, /k/, /ʃ/, /ts/, /ϕ/. All of these consonants
3 may contribute to the high vowel devoicing environment, when they occur as C₁ in C₁V_[High]C₂
4 sequences, but some of them are also claimed to condition deletion—not just devoicing—of the
5 following vowel in the same environment. According to Kawakami (1977), /s/, /ts/, and /ϕ/
6 condition deletion of the following /u/, while /k/ and /ʃ/ condition devoicing only, although
7 recall that he offers no phonological or phonetic evidence. According to Whang (2014), vowel
8 deletion occurs when the identity of the devoiced vowel is predictable from context. His
9 recoverability-based theory predicts that /u/ will be deleted following /s/, /ts/, /ϕ/, and /k/ but
10 that /u/ will be present (though devoiced) following /ʃ/. The predictions match Kawakami’s
11 intuition for four of the five consonants in our stimuli (/s/, /ts/, /ϕ/, /ʃ/). The point of divergence
12 is the /k/ environment. Whang’s theory predicts vowel deletion following /k/, whereas
13 Kawakami claims that the vowel is present (although devoiced) following /k/. The predictions
14 for the stimulus set from Whang (2014) are labelled as “(W)” in the first column of Table 1;
15 those due to Kawakami are labelled “(K)”; converging predictions are labelled as “(W,K)”.
16
17 We did not include stimuli in which high vowels are surrounded by two sibilants, as it is
18 known that devoicing may be inhibited in this environment (Hirayama 2009; Fujimoto 2015;
19 Maekawa & Kikuchi 2005; Tsuchida 1997). In addition, if the vowel is followed by allophones

1 of /h/ (including ϕ), devoicing may be inhibited (Fujimoto 2015). Our stimuli avoided this
2 environment as well.

3
4 We avoided any words in which the vowel following the target vowel is also high, because
5 consecutive devoicing is variable (Fujimoto 2015; Nielson 2015). We also chose near minimal
6 pairs in such a way that accent always matches within a pair. More specifically, /u/s in /hakusai/
7 and /yakuzai/ are both accented, but all the other target /u/s are unaccented. Tsuchida (1997)
8 shows that young speakers, at the time of 1997, show no effects of pitch accent on devoicing,
9 so that controlling for accent is important but may not be crucial.³ All the stimulus words are
10 common words.

11
12 Each participant produced 10-15 repetitions of the 10 target words in the carrier phrase: *okee*
13 _____ *to itte* ‘Okay say _____’. The preceding word *okee* was chosen, as it ends with a vowel
14 /e/, which differs both in height and backness from the target vowel, /u/. Participants were
15 instructed to speak as if they were making a request of a friend.

16
17 Each target word was produced 10-15 times by each speaker, generating a corpus of 690
18 tokens for analysis. Words were presented in Japanese script (composed of hiragana, katakana

³ There is very little if any durational differences between accented and unaccented vowels (Beckman, 1986), which would otherwise potentially affect the detectability of /u/. Since accented /u/ is not longer than unaccented /u/, this is yet another reason not to be too concerned about the placement of accent.

and kanji characters as required for natural presentation) and fully randomized with 10 additional filler items that did not contain /u/.

Equipment

The current experiment used an NDI Wave electromagnetic articulograph system sampling at 100 Hz to capture articulatory movement. NDI wave 5DoF sensors were attached to three locations on the sagittal midline of the tongue, and on the upper and lower lips near the vermillion border, lower jaw (below the incisor), nasion and left/right mastoids. The most anterior sensor on the tongue, henceforth TT, was attached less than one cm from the tongue tip. The most posterior sensor, henceforth TD, was attached as far back as was comfortable for the participant, ~4.5-6 cm. A third sensor, henceforth TB, was placed on the tongue body roughly equidistant between the TT and TD sensors. Acoustic data were recorded simultaneously at 22 KHz with a Schoeps MK 41S supercardioid microphone (with Schoeps CMC 6 Ug power module).

Stimulus display

Words were displayed on a monitor positioned 25cm outside of the NDI Wave magnetic field. Stimulus display was controlled manually using an Eprime script. This allowed for online monitoring of hesitations, mispronunciations and disfluencies. These were rare, but when they occurred, the experimenter repeated the trial. Each trial consisted of short (500 ms) preview presentation of the target word followed by presentation of the target word within the carrier phrase. The purpose of the preview presentation was to facilitate fluent reading of the target word within the carrier phrase, since it is known that brief presentation facilitates planning

(Davis et al., 2015) and, in particular, to discourage insertion of phonological phrase boundary between “okee (okey)” in the carrier phrase and the target word.

Post-processing

Following the main recording session, we also recorded the occlusal plane of each participant by having them hold a rigid object, with three 5DoF sensors attached to it, between their teeth. Head movements were corrected computationally after data collection with reference to three sensors on the head, left/right mastoid and nasion sensors, and the three sensors on the occlusal plane. The head-corrected data was rotated so that the origin of the spatial coordinates corresponds to the occlusal plane at the front teeth.

Analysis

Presence of voicing

Two authors went through the spectrograms and waveforms of all the tokens, and confirmed that /u/ in the devoicing environments are all devoiced (Figure 4 below provides a sample spectrogram), whereas /u/ in the voicing environments were voiced. This is an unsurprising result, given that vowel devoicing is reported to be obligatory in the normal speech style of Tokyo Japanese speakers (Fujimoto 2015).⁴

⁴ Though see Maekawa & Kikuchi (2005) who show that devoicing may not be entirely obligatory in spontaneous speech—in their study, overall, /u/ is devoiced about 84% of the time in the devoicing environment. However, as Hirayama (2009) points out, their study is likely to

Lingual targets

All stimulus items were selected so that the vowels preceding and following the target /u/ were non-high. In order to progress from a non-high vowel to /u/, the tongue body must rise. For some stimulus items, e.g., *ϕ_usoku~ϕ_uzoku*, and *f_utaisei~f_udaika*, the tongue body may also retract from the front position required for /e/ in the carrier phrase (*okee _____ to itte*) to the more anterior position required for /u/. The degree to which the tongue body retracts for /u/, i.e., the degree to which /u/ is a back vowel, has been called into question, with some data suggesting that /u/ in Japanese is central (Nogita, Yamane, & Bird, 2013), similar to “fronted” variants of /u/ in some dialects of English (e.g., Blackwood-Ximenes, Shaw, & Carignan, submitted; Harrington, Kleber, & Reubold, 2008). We therefore focus on the height dimension, in which /u/ is uncontroversially distinct from /o/. As an index of tongue body height, we used the TD sensor, the most posterior sensor of the three sensors on the tongue, which provides comparable data to past work on vowel articulation (Browman & Goldstein, 1992a; Johnson, Ladefoged, & Lindau, 1993)

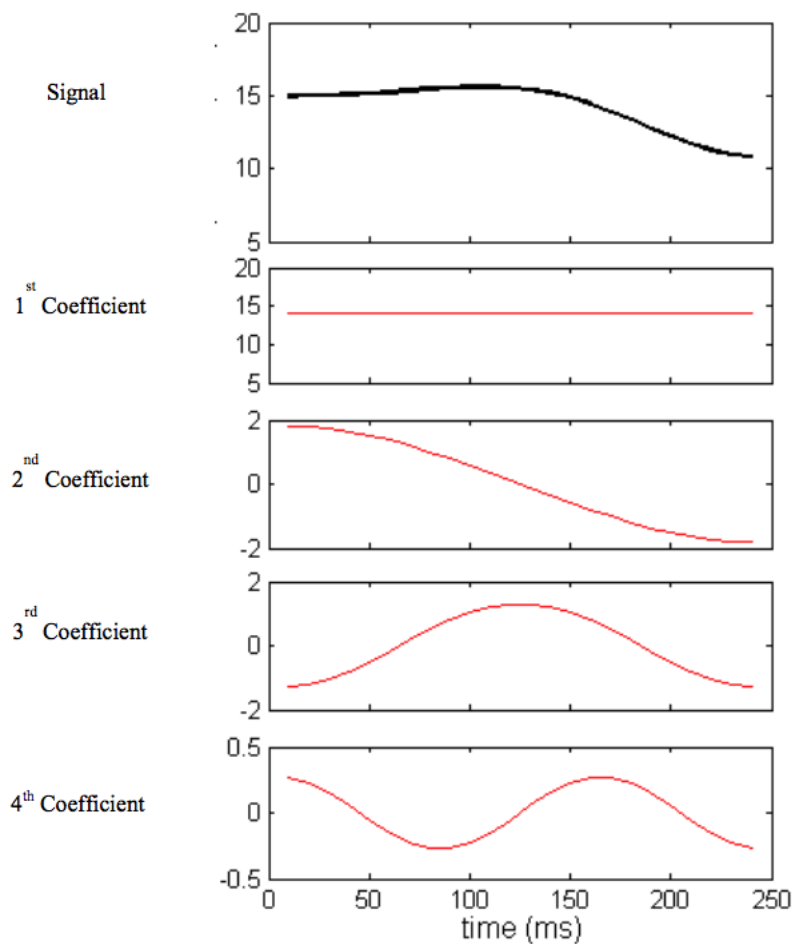
Our analytical framework makes use of the computational toolkit for assessing phonological specification proposed by Shaw and Kawahara (submitted). This framework evaluates presence/absence of an articulatory target based upon analysis of continuous movement of the tongue body across V1C1uC2V3 sequences (Figure 2). Analysis involves four steps: (1) fit Discrete Cosine Transform (DCT) components to the trajectories of interest; (2) define the

contain environments where there are two consecutive high vowels, which sometimes resist devoicing (Kondo 2001; Nielsen 2015).

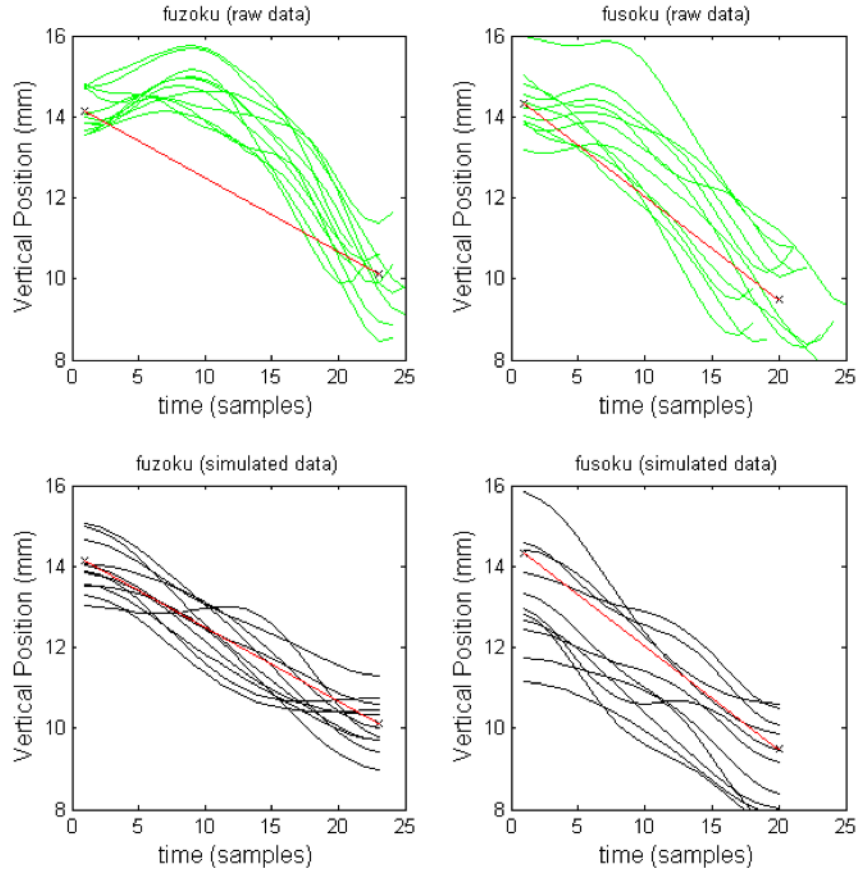
targetless hypothesis based on linear movement trajectories between the vowels flanking /u/;
(3) simulate “noisy” linear trajectories using variability observed in the data, in which the means
are taken from the DCT coefficients of the targetless trajectory, and the standard deviations are
taken from the observed data; (4) classify voiceless tokens as either “vowel present” or “vowel
absent” based on comparison to the “vowel present” training data (voiced vowels) and the
“vowel absent” training data (based on linear interpolation), using a Bayesian classifier.

**Figure 2: Steps illustrating the computation analysis (based on Shaw & Kawahara,
submitted)**

**(A) Step 1: Fitting four DCT components to a trajectory. The top panel is the raw signal
of /V1C1uC2V3/. The rest of the panels shows each DCT components.**



(B) Steps 2 & 3: Define a linear interpolation (shown as red lines) and generate a noisy null trajectory (the bottom two panels), based on the variability found in the raw data (the top two panels). Left=trajectories for /eφuzo(ku)/; right=trajectories for /eφuso(ku)/.



(C) Step 4: Train a Bayesian classifier in terms of DCT coefficients. “Vowel present” defined in terms of voiced vowels. “Vowel absent” defined in terms of the linear trajectory.

$$p(T|Co_1, Co_2, Co_3, Co_4) = \frac{p(T) p(Co_1, Co_2, Co_3, Co_4|T)}{p(Co_1, Co_2, Co_3, Co_4)}$$

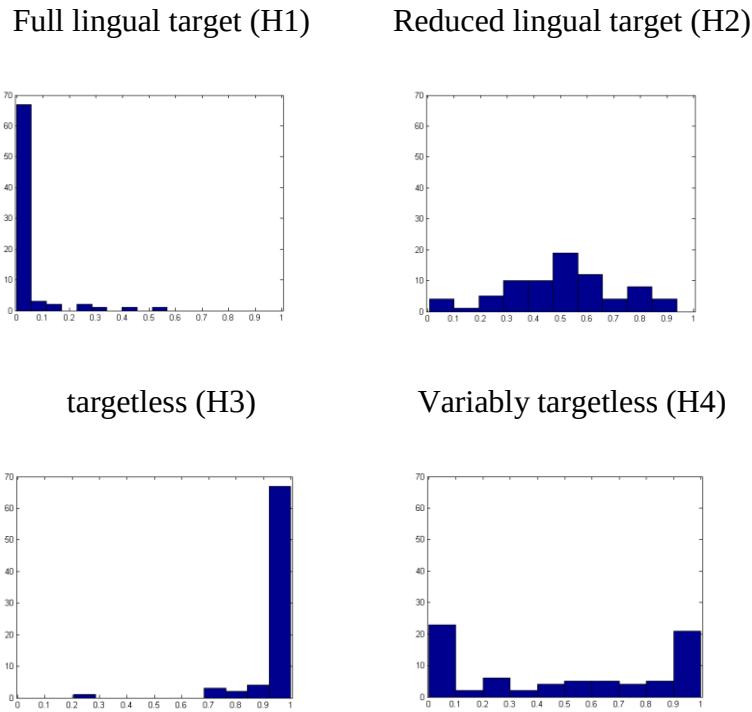
where

T = targetless, target present
*Co*₁ = 1st DCT Coefficient
*Co*₂ = 2nd DCT Coefficient
*Co*₃ = 3rd DCT Coefficient
*Co*₄ = 4th DCT Coefficient

Shaw and Kawahara (submitted) demonstrate that four DCT parameters are sufficient for representing TD height trajectories over VC_uCV intervals with an extremely high degree of precision ($R^2 > .99$). They show moreover that each DCT coefficient is linguistically interpretable: the 1st coefficient represents general TD height, the 2nd coefficient represents V1-V3 movement, the 3rd coefficient represents the presence of /u/, if any, and the 4th coefficient represents coarticulatory effects from surrounding consonants. We report the classification results for each devoiced token in terms of the posterior probability of targetlessness, i.e. the likelihood that the trajectory follows a linear interpolation between V1 and V3 instead of rising like the tongue body does in voiced tokens of /u/.

The four hypotheses introduced in (1) can each be evaluated by examining the distribution of posterior probabilities across tokens. Figure 3 presents hypothetical distributions corresponding to each of the hypotheses. If devoiced vowels have full lingual targets (H1), then we expect the probability of targetless to be low, as is shown in the top left panel of Figure 3. If H2 is correct, and devoiced vowels have reduced lingual targets, then the distribution of posterior probabilities should be centered around .5, as in the top left panel. If devoiced vowels lack lingual targets altogether (H3), then the distribution of posterior probabilities should be near 1.0, as in the bottom left panel of the figure. Lastly, if lingual targets are variably present, as in H4, then we expect to see a bimodal distribution, with one mode near 0 and the other near 1.0, as shown in the bottom right panel.

Figure 3: Four hypothetical posterior probability patterns. The vertical axis of each histogram shows posterior probabilities generated by the Bayesian classifier summarized in Figure 2(C). The histogram in the top left panel was obtained by submitting the /ϕuzoku/ tokens to the Bayesian classifier. The histogram in the bottom left was obtained by submitting the same number of simulated “vowel absent” trajectories to the classifier. The top right panel was generated by stochastic sampling of DCT coefficients that were averaged between “target present” (H1) and “target absent” (H3) values. The right bottom panel was created by averaging the targetless and the full vowel target tokens.



Posterior probabilities of targetlessness

Consonantal timing across devoiced vowels

In addition to examining the continuous trajectory of the tongue body, we also investigated the timing of the ballistic movements of consonants with TD movement trajectory. If the flanking consonants are coordinated in time with the lingual gesture for the vowel (e.g., see Smith, 1995 for a concrete proposal), reduction or deletion of that gesture may perturb consonant timing. In this way, consonant timing offers another independent measure of how devoicing influences lingual articulation.

For this analysis, we parsed articulatory landmarks from consonants flanking /u/ based on the primary oral articulator, e.g., tongue tip for /t/, /s/, tongue blade for /ʃ/, tongue dorsum for /k/, lips for /p/, etc, for each gesture. We determined the start and end of consonantal gestures with reference to the velocity signal in the movements and toward away from constriction. Both landmarks were extracted at the timepoint corresponding to 20% of peak velocity in the movement towards/away from consonantal constrictions, a heuristic applied extensively in work on consonant cluster timing (Bombien, Mooshammer, & Hoole, 2013; Gafos, Hoole, Roon, & Zeroual, 2010; Marin, 2013; Marin & Pouplier, 2010; Shaw, Gafos, Hoole, & Zeroual, 2011; Shaw, Gafos, Hoole, & Zeroual, 2009). For the sake of illustration, Figure 4 labels these landmarks for /ʃ/ and /t/, the consonants flanking target /u/ in /ʃutaisei/. The vertical black lines indicate the timestamp of the consonantal landmarks. They extend from the threshold of the velocity peak used as a heuristic for parsing the consonant up to the corresponding positional signal. The dotted line in the velocity panels extends from 0 (cm/s), or minimum velocity. For simplicity in display the figure shows only the vertical position and corresponding velocity but, whenever practicable all three dimensions of the position signal and corresponding tangential velocities were used to parse consonantal gestures. The interval between landmarks, labeled C1

for /ʃ/ and C2 for /t/, are the consonant durations. The interval between the consonants, or inter-consonantal interval (ICI), defined as the achievement of target of C2 minus the release of C1 (see also, inter-plateau interval in Shaw & Gafos, 2015) was also analyzed. At issue is whether this interval varies with properties of the lingual gesture for the intervening /u/.

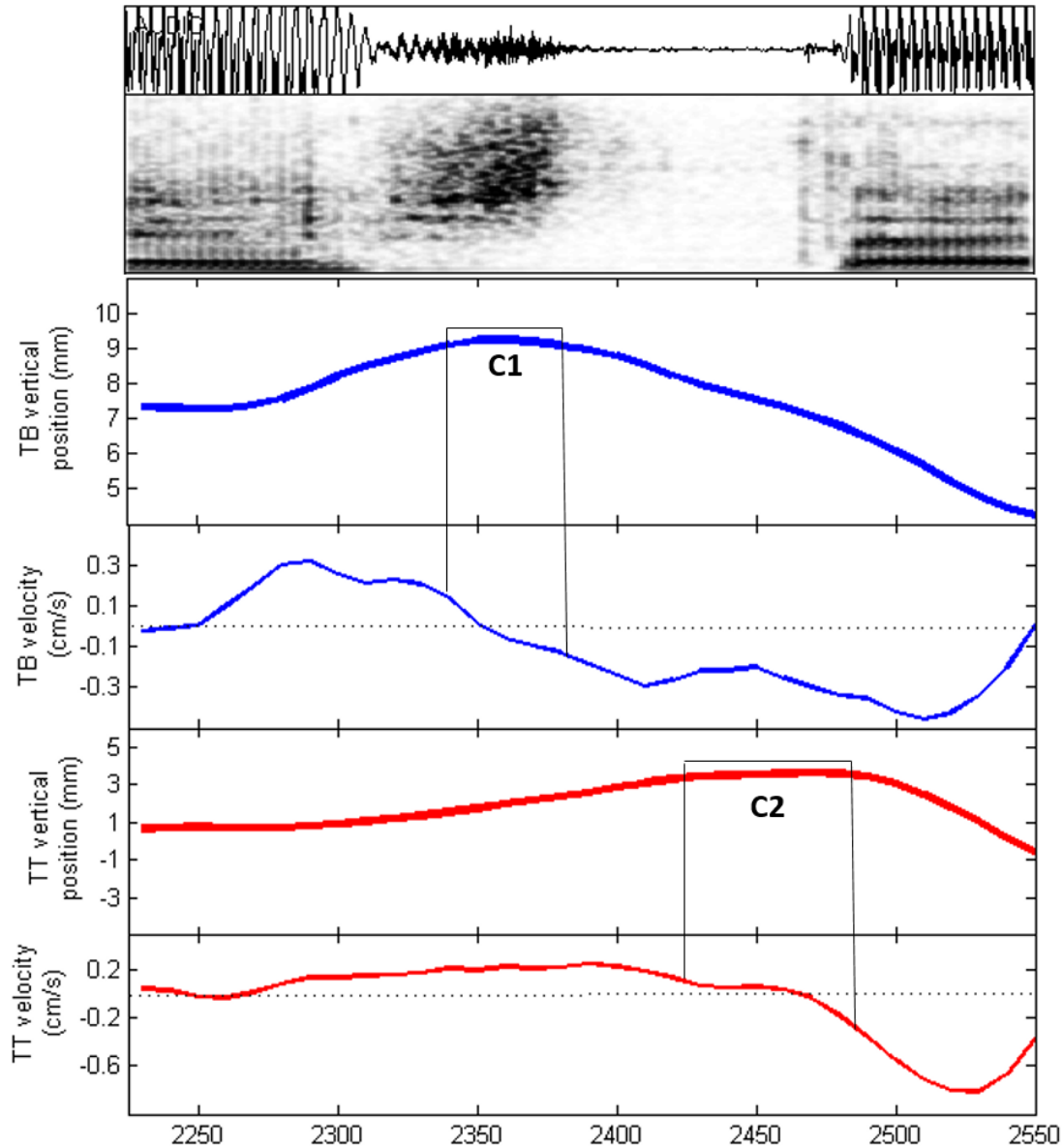


Figure 4: An illustration of how consonantal landmarks, /ʃ/ (C1) and /t/ (C2), were parsed from a token of /ʃuʔaisei/. The thick blue line shows the vertical position of the

tongue blade (TB); the thin blue line shows the corresponding velocity signal. The thick red line shows the vertical position of the TT; the thin red line shows the corresponding velocity signal. Consonant onsets and offsets were based on a threshold of peak velocity, vertical black lines, in the movements toward and away from target.

Data exclusion

The tongue tip sensor became unresponsive for S04 on the sixth trial. We think that this was due to wire malfunction, possibly due to the participant biting on the wire. The tongue tip trajectory is relevant only for the analysis of consonant timing. Due to missing data, the consonant timing analysis for this speaker is based on only 5 trials. Analyses involving other trajectories are based on 11 trials for this participant.

Results

Presence/absence of articulatory height targets

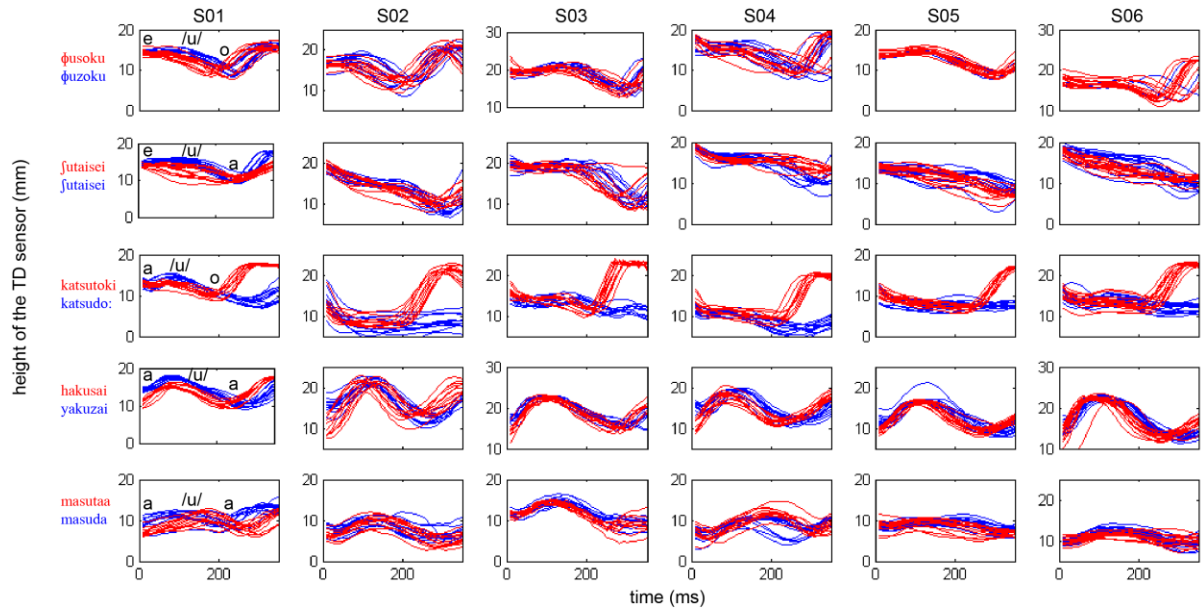
Figure 5 summarizes the data on TD height across speakers and words. Each panel shows TD height (y-axis) over time (x-axis) for tokens of a target word with a voiced vowel (blue lines) and devoiced counterpart (red lines). The columns show data from different speakers and the rows show the different voiced-devoiced dyads. An interval of 350ms (35 samples of data), beginning with the vowel preceding the target, is shown in each panel. Despite the variation in speech rate (note, for example, that the rise of the TD for /k/ at the right of panels displaying /ɸusoku/~/ɸuzoku/ in the top row is present to different degrees across speakers), a 350ms window is sufficient to capture the three-vowel sequence including the target /u/ and preceding and following vowels for all tokens of all words across all speakers.

To facilitate a visual parse of the tongue height trajectories, annotations are provided for the first speaker (leftmost column of panels). The trajectories begin with the vowel preceding the target vowel, e.g., /e/ from the carrier phrase in the case of /ϕusoku/ and /ʃutaisei/, /a/ in /kʌtutoki/, etc. Movements corresponding to the vowel following /u/ are easy to identify—since the vowel following /u/ is always non-high, it corresponds to a lowering of the tongue. The label for the target vowel, /u/, has been placed in slashes between the flanking vowels. We note also that the vertical scales have been optimized to display the data on a panel-by-panel basis and are therefore not identical across all panels. In particular, across speakers the TD is lower in the production of *masuda~masutaa* than for many of the other dyads and that this influences the scale for most speakers (S02-S06).

Differences between voiced and devoiced dyads (red and blue lines, respectively, in the figure) include cases in which the tongue tends to be higher in the neighborhood of /u/ for the voiced than for the devoiced member of the dyad. The top left panel, /ϕusoku/ for S01, exemplifies this pattern (a zoom-in Figure provided in Figure 6). In this panel, the blue lines rise from /e/ to /u/ while the devoiced vowel trajectory is a roughly linear trajectory between /e/ and /o/. Our analysis assesses this possibility specifically by setting up stochastic generators of the competing hypothesis, lingual target present (based on the voiced vowel) vs. lingual target absent (based on linear interpolation), and evaluates the degree to which the voiceless vowel trajectory is consistent with these options.

1

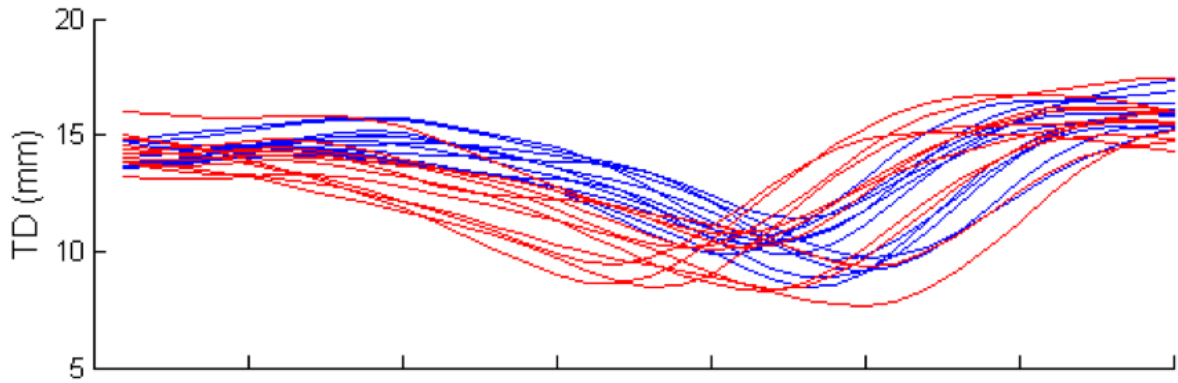
Figure 5: TD trajectories of all speakers, all items.



2

3

Figure 6: TD trajectories of /ɸusoku/ and /ɸuzoku/ for Sp01.



4

5

6 Table 2 provides the average posterior probabilities across tokens by speaker and by word. The
7 probability of targetlessness varies across speakers rather widely, from speakers that have a
8 high probability of targetless vowels, e.g. 0.66 for S04, to speakers with a much lower
9 probability of targetlessness, e.g., 0.17 for S05. Within speakers, there are some clear patterns
10 across words. All speakers have a higher probability of targetlessness in /ɸutaisei/ than in

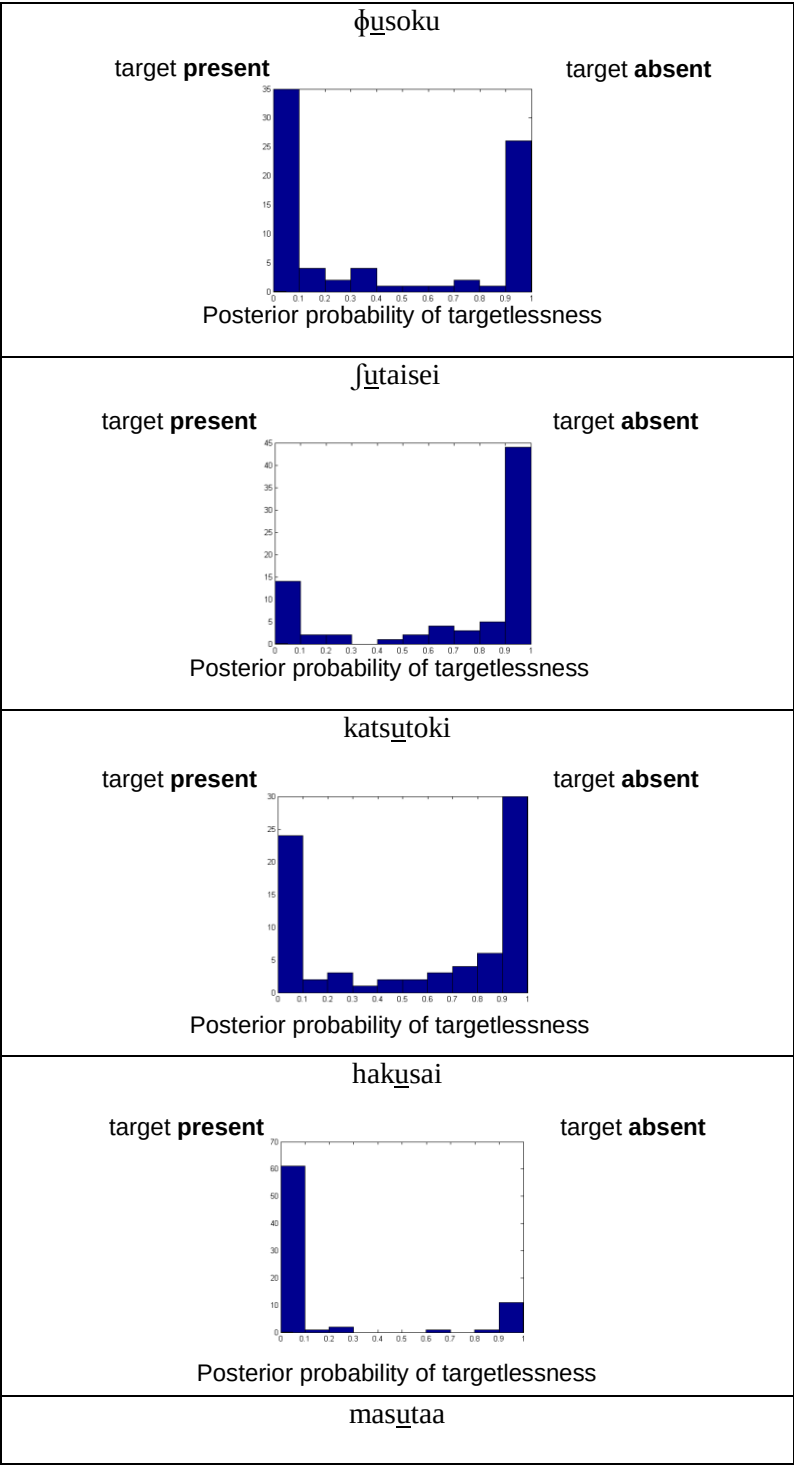
/ɸ_usoku/ and a higher probability of targetlessness in /ɸ_usoku/ than in /hak_usai/. Expressed in terms of the consonantal contact that results from a targetless /u/, the pattern across speakers is $\int_t > \phi_s > k_s$, where “>” indicates greater degree of targetlessness. How the other two words fit in with this pattern varies across speakers. For all speakers, the /u/ in /kats_utoki/ has a higher probability of targetlessness than /hak_usai/, but speakers vary in how target probability in /hak_usai/ relates to /ɸ_usoku/ and /ɸ_utaisei/. Lastly, lingual target probability in /mas_utaa/ is variable both across subjects and also within-subjects in terms of how it patterns with other words.

Table 2: Posterior probability of lingual targetlessness

	S01	S02	S03	S04	S05	S06	average
ɸ _u soku	0.47	0.39	0.75	0.84	0.01	0.19	0.44
ɸ _u taisei	0.92	0.68	0.84	0.99	0.02	0.89	0.72
kats _u toki	0.81	0.19	0.69	0.93	0.06	0.79	0.58
hak _u sai	0.00	0.00	0.51	0.50	0.00	0.07	0.18
mas _u taa	0.64	0.09	0.01	0.02	0.74	0.41	0.32
average	0.57	0.27	0.56	0.66	0.17	0.47	0.45

The histograms of posterior probabilities in Figure 7 provide a direct test of our hypotheses. For all words, the distribution of probabilities is distinctly bimodal. Most of the tokens of devoiced vowels produced by our 6 speakers were either clearly produced with a lingual height target or clearly produced without one. Only a small number of tokens are intermediate. Given the distribution of posterior probabilities, we can safely interpret the average probabilities in the table as tendencies to either (1) produce a vowel height target or (2) proceed linearly from

- 1 the V1 target to target the V3 target. These results support the “optional targetless hypothesis”
2 (=H4) (see also Figure 3).
3



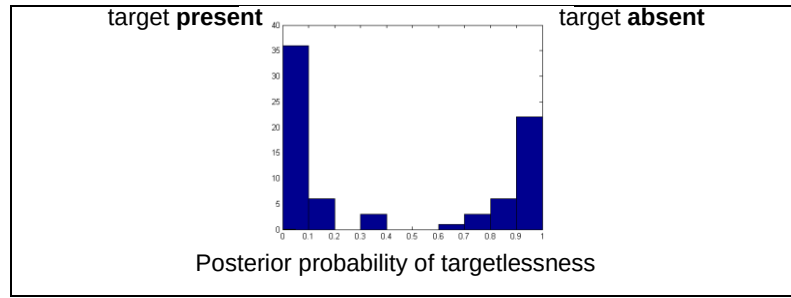


Figure 7: Posterior probabilities of targetlessness for each stimulus.

Consonantal timing across targetless vowels

We next turn to the timing of the consonants flanking /u/, asking whether the absence of the lingual target influences relative timing between the preceding and following consonants. Figure 8 summarizes the inter-consonantal interval (ICI), defined as the interval spanning from the release of C1 to the achievement of target of C2. The ICI is plotted for tokens containing a vowel target and tokens in which the vowel target was absent. Since ICI is a difference, it is possible for it to be negative. This can happen when C2 achieves its target before the release of C1, as commonly observed in English consonant clusters (e.g., Byrd, 1996). The average ICI is around zero for $\phi_{\underline{u}}soku \sim \phi_{\underline{u}}zoku$ indicating that the lips remain approximated until the tongue blade achieves its target, a temporal configuration which does not at all interfere with the achievement of a height target for the vowel. The / ϕs / sequence of consonants is a front-to-back sequence, in that the place of articulation for the labial fricative, the first consonant, is anterior to the place of articulation of the alveolar fricative, the second consonant. Consonant clusters with a front-to-back order of place tend to have increased overlap in consonant clusters than back-to-front clusters (Chitoran, Goldstein, & Byrd, 2002; Gafos et al., 2010; Wright, 1996; Yip, 2013). For other dyads, the consonantal context requires longer average ICI's, with median values ranging from ~50 ms to ~170 ms. The variation reflects the broader fact about Japanese that consonantal context has a substantial influence on the duration of the following vowel (see Kawahara & Shaw, submitted for a large scale corpus study).

Our primary interest in the ICI interval is whether it is impacted by the presence or absence of a lingual target for the intervening vowel. We opted to analyze the data in terms of this categorical difference (instead of using the raw probabilities as predictors) because, as shown in the histograms in Figure 7, the data are largely categorical in nature. We applied the Bayesian decision rule, interpreting (targetless) probabilities greater than .5 as indicating that the vowel height target was absent and probabilities less than .5 as indicate that the target was present. Figure 8 shows that the effect of lingual targetlessness on ICI varies across consonantal environments. For *futaisei~fudaika* and *katsutoki~katsudou*, ICI is longer when the vowel is present. For the other three dyads, *φusoku~φuzoku*, *hakusai~yakuzai*, and *masutaa~masuda*, the presence of a lingual vowel target actually results in shorter ICI.

To assess the statistical significance of the trends shown in Figure 8, we fitted a series of three nested linear mixed effects models to the ICI data using the *lme4* package (Bates, Maechler, Bolker, & Walker, 2014) in R. The baseline model included word dyad as a fixed factor, since it is clear that ICI depends in part on the identity of the particular consonants, and random intercepts for speaker, to account for speaker-specific influences on ICI. The second model added to the baseline model a fixed factor, the presence vs. absence of a vowel height target, henceforth TARGET PRESENCE. The third model included the interaction between word dyad and the TARGET PRESENCE. Table 3 summarizes the model comparison. Adding TARGET PRESENCE to the model leads to only marginal improvement over the baseline. This is because TARGET PRESENCE impacts ICI for some dyads positively and for other dyads negatively. Adding the interaction between dyad and TARGET PRESENCE leads to a significant improvement over the model with dyad and TARGET PRESENCE as non-interacting fixed factors. These results indicate that the trends observed in Figure 8 are statistically reliable. ICI varies depending on whether

there is a vowel height target between consonants and how it varies depends on the identity of the flanking consonants.

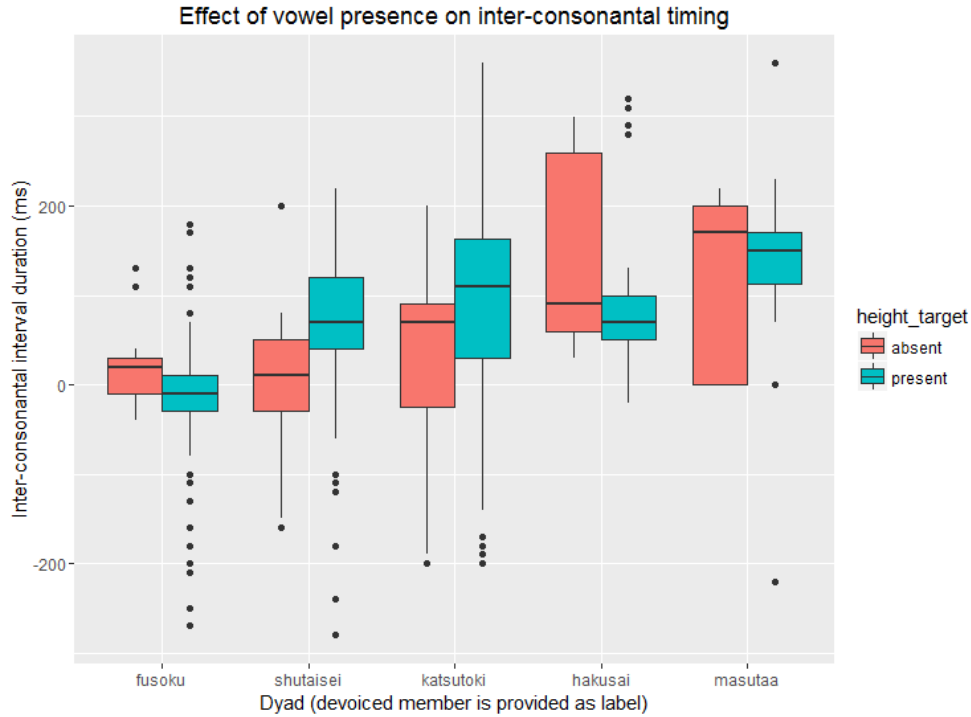


Figure 8: Inter-consonantal interval duration for each item, classified by target absence/presence.

Table 3: Model comparison for ICI duration.

Model of ICI	Df	AIC	BIC	logLik	Chisq	Pr(>Chisq)
dyad + (1 speaker)	7	7869	7901	-3928	-	-
dyad + target_presence + (1 speaker)	8	7868	7904	-3926	3.72	0.054
dyad* target_presence + (1 speaker)	12	7838	7892	-3907	38.06	0.0000001***

We also investigated how TARGET PRESENCE influences the duration of C1 and C2. The data for C1 are shown, again by dyad, in Figure 9.

For four of the five dyads—the exception is *ϕusoku~ϕuzoku*—C1 is somewhat longer when it is not followed by a vowel height target. The largest effects are for *futaisei~fudaika* and for *katsutoki~katsudou*. The effect of TARGET PRESENCE on C1 duration for these dyads is in the opposite direction as on ICI. We return to this negative correlation in the discussion below. Table 4 shows the statistical results for C1 duration, again based on comparison of nested models. The main effect of TARGET PRESENCE was significant, indicating a reliable overall trend for C1 to be longer when the vowel target is absent for the following vowel. The significant interaction between TARGET PRESENCE and DYAD indicates that the magnitude of the TARGET PRESENCE effect on C1 is not uniform across dyads.

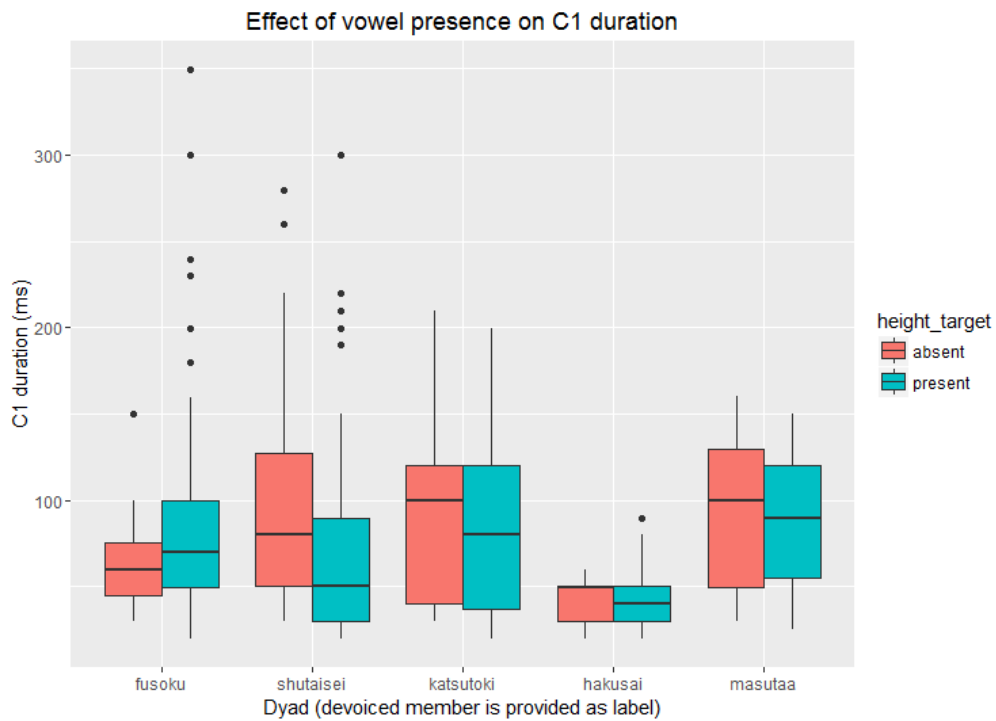


Figure 9: C1 duration for each item.

Table 4: Model comparison for C1 duration.

Model of C1 duration	Df	AIC	BIC	logLik	Chisq	Pr(>Chisq)
dyad + (1 speaker)	7	6956	6987	-3471		
dyad + target_presence + (1 speaker)	8	6954	6990	-3569	4.04	0.04*
dyad* target_presence + (1 speaker)	12	6945	6999	-3461	16.39	0.003**

The results for C2 duration are shown in Figure 10. For most dyads, C2 is longer when the vowel target is absent, although the difference in degree varies substantially across dyads. Again, the largest effects are for *futaisei~fudaika* and *katsutoki~katsudou*. As shown in Table 5, both the main effect of TARGET PRESENCE and the interaction with DYAD are statistically significant.

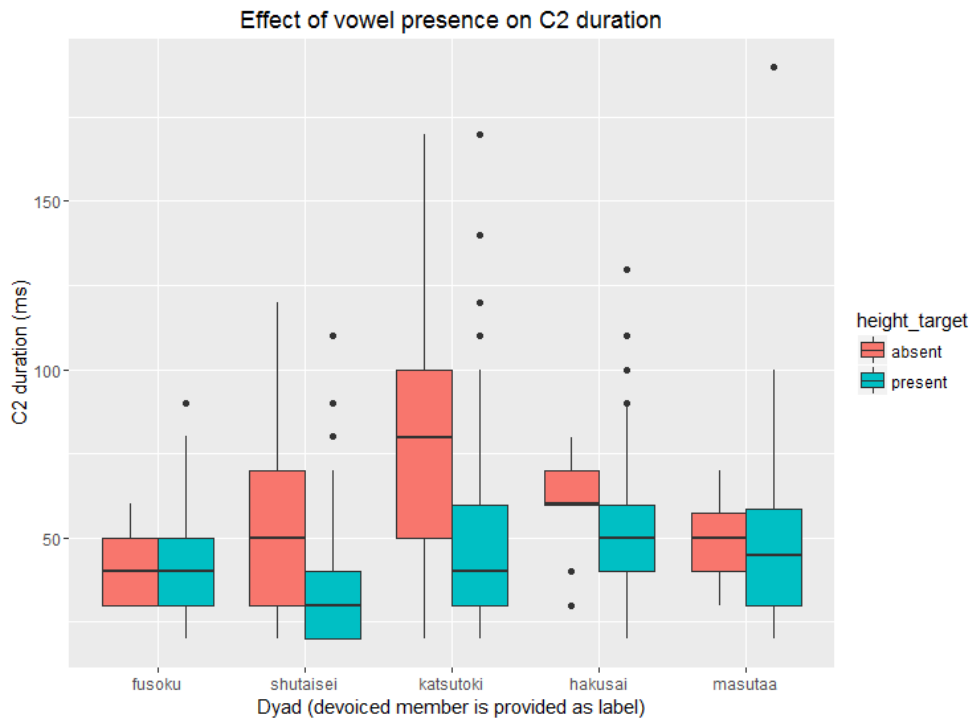


Figure 10: C2 duration.

Table 5: Model comparison.

Model of C2 duration	Df	AIC	BIC	logLik	Chisq	Pr(>Chisq)
dyad + (1 speaker)	7	6104	6135	-3045		
dyad + target_presence + (1 speaker)	8	6070	6106	-3027	35.47	0.0000000009***
dyad * target_presence + (1 speaker)	12	6042	6096	-3009	35.68	0.0000003**

To summarize the consonant duration results, *futaisei~fudaika* and *katsutoki~katsudou* patterned differently from the other dyads. The pattern observed for these two dyads is that, when /u/ is targetless, C1 and C2 lengthen while ICI shortens. The tradeoff in duration between C1 and ICI is reminiscent of reports of acoustic data showing that vowels have shorter duration when followed by consonants with longer duration, i.e., voiceless consonants, than when followed by consonants with shorter duration, i.e., voiced consonants (Kawahara & Shaw, submitted; Port, Al-Ani, & Maeda, 1980). Positive ICI values closely approximate the period of open vocal tract that corresponds to acoustic measures of vowels. A difference between the tradeoff between acoustic intervals observed in past studies is that our data show a tradeoff between consonant duration, measured articulatorily and the presence/absence of a lingual articulatory target in the following vowel. We note also that this tradeoff occurs in the two dyads which have the highest incidence of vowel targetlessness. The other dyads do not show the same, on average, tradeoff between C1 duration and ICI. The effect of vowel targetlessness on C2 duration is more uniform across dyads. C2, the consonant following /u/, tends to be longer when the preceding vowel is targetless.

To summarize, our results provide support for H4, the hypothesis that devoiced vowels in Japanese are optionally targetless. Speakers differ in the degree to which they produce words without a vowel target, but all speakers produce /u/ without a lingual target in some words some

of the time. The tendency to produce a targetless vowel varies systematically across words, probably due to the consonantal environment in which /u/ occurs. Targetlessness is most common in /ʃutaisei/ followed by /ɸusoku/ and then /hakusai/, a pattern which is consistent across all speakers. Vowel targetlessness also impacts the timing of flanking consonants. The duration of C1, C2, and the interval between them, ICI, were all influenced by the presence/absence of a vowel target, while the direction and degree of influence varied systematically across dyads.

Discussion

The major purpose of the study was to examine the lingual articulation of devoiced vowels in Japanese. At the beginning, we formulated four specific hypotheses, based on claims in the previous literature. The EMA experiment reported in the paper, analyzed via Bayesian classification, supports the hypothesis that lingual gestures are optionally deleted or, put more neutrally, that they are optionally present. Speakers varied in the frequency with which they produced /u/ without a lingual target, but there was systematicity among speakers in such a way that the vowel target was more likely in some environments than others.

Recall that there are two previous theories (Kawakami 1971; Whang 2014) that predicted deletion in certain environments. Comparing Table 1, which shows the predictions of these theories, and Table 2, which shows the actual deletion probabilities, our results do not match with either. In particular, both Kawakami (1971) and Whang (2014) argue for devoicing, not deletion, in [ʃutaisei], but we found the highest probability of targetlessness in that environment. Also, our results that all words have non-zero probabilities of targetlessness, suggesting an optional process of target specification (or deletion), although the probabilities differ across environments.

1 One theoretical consequence of our results is that they reveal a clear case of a categorical but
2 optional pattern. As recently pointed out by Bayles, Kaplan & Kaplan (2016), “optionality” in
3 many studies can come from averaging over inter-speaker differences, and it is important to
4 examine whether a categorical pattern can show true optionality *within a speaker*. Although
5 this was not the main purpose of this experiment, our finding supports the view expressed by
6 Bayles et al. (2016) that indeed, a categorical pattern, like French schwa deletion, can be
7 optional within a speaker. Revealing such patterns requires that the phonological status of each
8 token is assessed individually. In the case of devoiced vowels in Japanese, the average trajectory
9 of devoiced /u/ would point to the misleading conclusion that /u/ is reduced, because it averages
10 over “full target” and “no target” tokens (see Shaw & Kawahara, submitted for further details).

11
12 Determining whether a phonological process is categorical and optional faces challenges that
13 are less of a concern in other areas of linguistics, e.g., morphology, syntax, when it is often
14 more clear, even obvious, whether an element is present or absent. The continuity of phonetic
15 measurements makes it natural to consider the possibility that a gesture is categorically present
16 but reduced in magnitude or coordinated in time with other gestures such that the acoustic
17 consequences of the gesture are attenuated (Browman & Goldstein, 1992b; Iskarous,
18 McDonough, & Whalen, 2012; Jun, 1996; Jun & Beckman, 1993; Parrell & Narayanan, 2014).
19 Particularly with vowels, gradient and gradual phonetic shift is well-documented and is often
20 treated as the primary mechanism of variation (e.g., Labov, Ash, & Boberg, 2005; Wells, 1982
21 for comprehensive overviews). This underscores the importance of deploying rigorous methods
22 to support claims about the presence/absence of a gesture in the phonetic signal of which the
23 Japanese data offer a clear case. We emphasize at this point also, that our method is useful in
24 testing the general “phonetic underspecification analysis” (Keating, 1988). Many research has
25 argued that some segments lack particular phonetic target (Cohn, 1993; Keating, 1988;

Pierrehumbert & Beckman, 1988), but they have never addressed the question of how much linearity can be considered as actuation of actual linearity, because articulatory patterns are never completely linear. The current computational analysis thus offers a general toolkit which can be used to address the general issue of phonetic underspecification.

We also investigated how the presence/absence of the lingual gesture for the vowel influences the timing between flanking consonants. Tokens that lack a lingual vowel target may be expected to have flanking consonants closer together in time. This would follow from, e.g., the proposal that C2, the consonant following /u/, is timed directly to the lingual gesture of the preceding vowel (see Smith, 1995, for a specific proposal along these lines). This result was obtained only for two of the dyads, namely *futaisei~fudaika* and *katsutoki~katsudou*. Interestingly, these two dyads also showed the highest probability of vowel target absence. Increased incidence of /u/ lacking lingual articulatory gestures may be feeding tighter C-C coordination in /tst/ and /ʃt/ relative to the other clusters.

The patterns in the duration of flanking consonants provided independent evidence for the targetlessness of /u/. All three intervals investigated, C1 duration, C2 duration, and the ICI were significantly affected by vowel targetlessness. We also observed tradeoffs between some of these intervals, which we return to here. Figures 11 shows a scatterplot of C1 duration and ICI. The red triangles are tokens that contain a vowel target. The blue triangles represent targetless tokens. The red and blue lines are linear fits to the tokens with and without lingual vowel targets, according to our Bayesian classification.

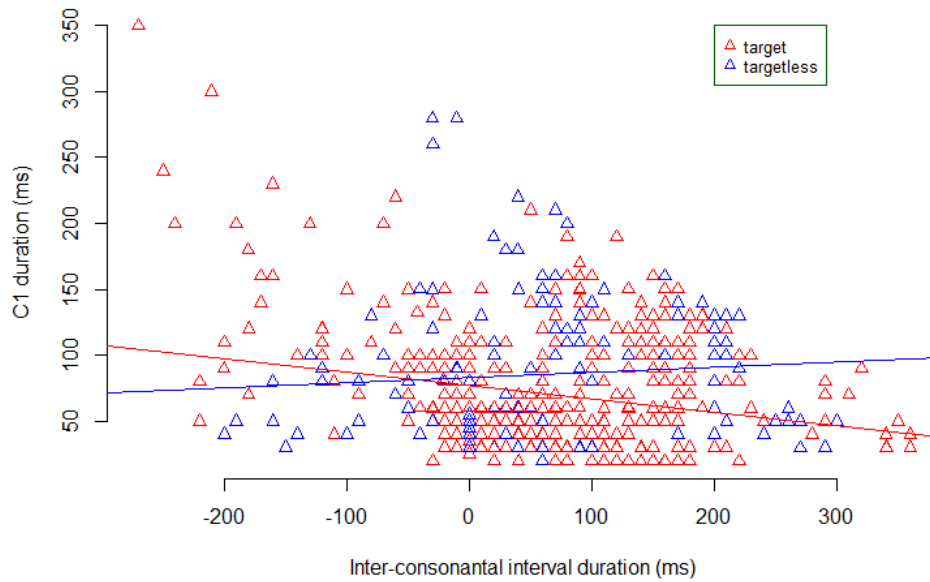
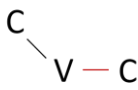
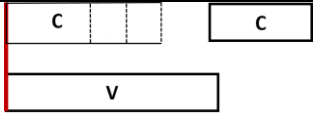


Figure 11: The correlation between C1 duration and ICI.

There is a significant negative correlation between C1 and ICI when the vowel target is present ($r(506) = -0.22$; $p < .001$). When absent, the relation between C1 and ICI is weakly positive and not statistically significant ($r(154) = 0.08$, $p = .32$). Thus, the tradeoff between consonant duration and ICI duration is only maintained when the lingual vowel target is present. We take this result as converging evidence for vowel targetlessness in those tokens classified as linear interpolations between flanking gestures. It may also suggest the emergence of a distinct pattern of consonant coordination in tokens that lack a lingual gesture for vowels, i.e., C-V coordination may be giving rise to C-C coordination. The comparison between coordination topologies, in the sense of Gafos (2002), and consequences for relative timing are summarized in Table 6. The rectangles represent activation durations for gestures. The dotted lines indicate variation in intra-gestural activation duration. Under C-V coordination, we assume here that the start of consonant and vowel gestures are coordinated in time, i.e., the gestures are in-phase (Goldstein, Nam, Saltzman, & Chitoran, 2009). Under this coordination regime, shortening of C1 would

expose more of the ICI interval, predicting the negative correlation between C1 duration and ICI that we observed for tokens that contain a lingual vowel gesture. Under C-C coordination, on the other hand, the end of C1 is coordinated with the start of C2. Variation in activation duration for C1 impacts directly when in time C2 begins. Thus, under C-C coordination no trade-off between C1 duration and ICI is predicted. This is the result observed when the lingual gesture of the vowel was absent.

Table 6: Schematic illustration of C1 duration variation under different coordination topologies.

	C-V coordination	C-C coordination
Coordination Topology		
Temporal intervals		

The variable targetlessness of devoiced /u/ raises several new research questions. Firstly, we have observed that targetless probabilities differ across words rather systematically. Even as the probabilities of targetlessness differed across speakers, the relative frequency of targetlessness across words was consistent within speakers. For all speakers, vowels were more likely to be targetless following *futaisei* than *katsutoki* and more likely to be targetless following *katsutoki* than *φusoku*. What conditions these systematic differences? The identity of the flanking consonants, the lexical statistics of the target words, the informativity of the vowel, etc., are all possibilities. Due to the small number of words recorded in this experiment we hesitate to speculate on which of these factors (and to what extent) may influence targetlessness,

1 but we plan to follow up with another study that expands the number of words instantiating the
2 consonantal environments reported here. A second question is the syllabic status of consonant
3 clusters that result from targetless /u/. Matsui (2014) speculates that the preceding consonant
4 forms an independent syllable, a syllabic consonant, while Kondo (1997) argues that it is
5 resyllabified into the following syllable, forming a complex onset. A third question is the status
6 of /i/, the other of the two high vowels that are categorically devoiced in Tokyo Japanese. Our
7 conclusion thus far is solely about /u/, which is independently known to be variable in its
8 duration (Kawahara & Shaw, submitted), and is more susceptible to coarticulation than /i/
9 (Recasens & Espinosa, 2009). At this point we have nothing to say about whether devoiced /i/
10 may also be targetless in some contexts but hold this to be an interesting question for future
11 research. Finally, the observed shift from C-V to C-C coordination may bear on the broader
12 theoretical issue of how higher level structure, i.e., the mora in Japanese, relates to the temporal
13 organization of consonant and vowel gestures. The CV mora may be less determinate of
14 rhythmic patterns of Japanese than is sometimes assumed (Beckman 1982; Warner and Arai,
15 2001).

Conclusion

18 The current experiment was designed to address the question of whether devoiced /u/ in
19 Japanese is simply devoiced (Jun & Beckman 1993) or whether it is also targetless (Kondo
20 2001). Since previous studies on this topic have used acoustic data (Whang 2014) or
21 impressionistic argumentation (Kawakami 1977), we approached this issue by collecting
22 articulatory data. Using EMA, we recorded tongue dorsum trajectories in words containing
23 voiced and devoiced /u/. Some devoiced tokens showed a linear trajectory between flanking
24 vowels, indicating that there is no lingual articulatory target for the vowel to speak of. Other

1 devoiced vowel tokens had trajectories like voiced vowels. Noticeably absent were tokens that
2 were reduced, showing trajectories intermediate between the voiced vowels and linear
3 interpolation between flanking vowels. We conclude that /u/ is optionally targetless; i.e., there
4 is token-by-token variability in whether the lingual gesture is present or absent. The patterns of
5 covariation between C1 duration and ICI, the period of open vocal tract spanning from the
6 release of C1 to the target of C2, provided further support for this conclusion. For tokens
7 classified as containing a lingual target, there is a significant linear correlation between C1
8 duration and ICI, a correlation which is absent when the vowel lacks a lingual articulatory
9 target.

10
11 Achieving the above descriptive generalization—that devoiced vowels are optionally
12 targetless—required some methodological advancements, including analytical tools for
13 assessing phonological status on a token-by-token basis. We analyzed the data using Bayesian
14 classification of a compressed representation of the signal based on Discrete Cosine Transform
15 (following Shaw & Kawahara, submitted). The posterior probabilities of the classification
16 showed a bimodal distribution, supporting the conclusion that devoiced /u/ in Tokyo Japanese
17 is variably targetlessness.

18
19 Overall, establishing that devoiced /u/ tokens are sometimes targetless, in that they do not differ
20 from the linear interpolation of flanking gestures, answers a long-standing question in Japanese
21 phonetics/phonology while raising several new questions to follow-up in future research,
22 including the syllabic status of consonant clusters flanking a targetless vowel, the role of the
23 mora (or lack thereof) in Japanese timing, the phonological contexts that favor targetless /u/,
24 and whether other devoiced vowels, particularly /i/, may also be variably targetless.

Acknowledgements

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